

# FIT3155: Week 8 Lab Sheet

(Questions are based on Week 7 topic dealing with Fibonacci heaps.)

**Objectives:** Develop a working understanding of the structure and operations of Fibonacci heaps, leading to how they support efficient amortized time for priority queue operations.

## Conceptual exercises

1. You studied Dijkstra's algorithm in FIT2004. Suppose you are given a general heap data structure that supports the operations **extract-min** and **decrease-key**. Express the overall time complexity of Dijkstra's algorithm on a graph  $G(V, E, W)$  in terms of:

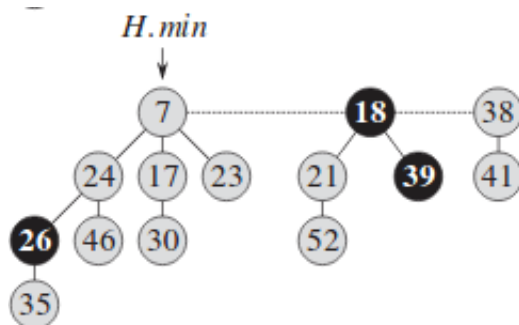
- the number of vertices  $|V|$ ,
- the number of edges  $|E|$ , and
- the time complexities of **extract-min** and **decrease-key** denoted by  $T_{\text{extract-min}}$  and  $T_{\text{decrease-key}}$ .

2. Insert the following elements into a Fibonacci heap:

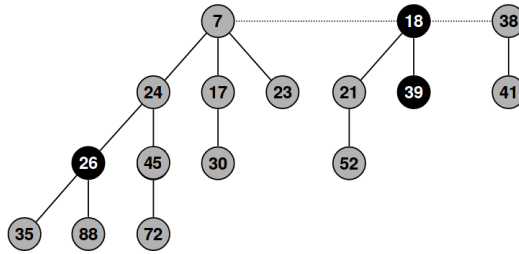
3, 8, 10, 11, 1, 12, 2, 17

and then perform **extract-min** on the resulting heap.

3. Perform **extract-min** on the following Fibonacci heap:



4. What are the worst-case time complexities of **insert** and **extract-min** operations?
5. Starting from the following state of a Fibonacci heap:



Run the following sequence of operations, and after each step, draw the resultant heap:

- (a) **decrease-key** of 45 to 40.
  - (b) **decrease-key** of 40 to 12.
  - (c) **decrease-key** of 35 to 1.
  - (d) **extract-min**.
6. Let  $F_n$  denote the  $n$ -th Fibonacci number, where  $F_n = F_{n-1} + F_{n-2}$ . Prove using induction that  $\sum_{i=0}^n F_i = F_{n+2} - 1$ .
  7. Using induction, show that for all integers  $n \geq 0$ , the  $(n+2)$ -th Fibonacci number satisfies  $F_{n+2} \geq \phi^n$ , where  $\phi = \frac{1+\sqrt{5}}{2}$  (aka, the Golden Ratio).
  8. Prove the maximum degree bound of  $O(\log N)$  on any node in a Fibonacci Heap containing  $N$  elements.

## Implementation exercise

(A message of caution: ‘Vibe coding’ with GenAI will hinder your learning of algorithms and data structures. The degree (and this unit) you have enrolled into is a specialist degree (unit). By definition it expects one to demonstrate specialist understanding of topics. This understanding comes from making a concerted effort, via thinking, implementing, debugging etc.)

1. Implement a Fibonacci heap data structure supporting the following operations:

- **insert**
- **min**
- **extract-min**
- **merge**
- **decrease-key**
- **delete**

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